

CognitiveOS

A Multi-Agent Framework for Human Memory, Personality, Emotion & Decision Modeling

Synopsis

CognitiveOS is a proposed multi-agent AI framework that models selected aspects of human cognition:

- long-term memory,
- personality traits,
- emotional state, and
- decision-making behavior.

The system is designed as a “cognitive layer” on top of modern large language models, where specialized agents handle memory storage and retrieval, personality representation, emotional appraisal, and reasoning. These agents interact to produce behavior that is more stable, human-like, and context-aware over time.

The goal of this project is twofold:

1. **Engineering Goal** – Build a working prototype that learns from a person’s text interactions (conversations, reflections, notes) and gradually forms a cognitive profile that influences future responses.
2. **Research Goal** – Study how integrating long-term memory, emotion, and personality inside a multi-agent architecture affects the stability, coherence, and human-likeness of the agent’s behavior. The intention is to produce a publishable paper based on the architecture, experiments, and evaluation.

Features (Planned Scope)

1. Memory Engine

- Episodic memory: stores significant “events” or interactions in chronological order.
- Semantic memory: stores factual knowledge and stable preferences.
- Memory compression: summarizes older information to keep it useful but compact.

- Contextual retrieval: fetches relevant past experiences for the current query.

2. Personality Engine

- Personality representation using a trait model (e.g., Big Five).
- Personality-conditioned prompting for the underlying LLM.
- Slow, controlled personality drift based on long-term experiences.

3. Emotion & Mood Engine

- Emotion appraisal based on events and user feedback (e.g., joy, sadness, fear, frustration).
- Short-term emotional state and longer-term mood.
- Emotional influence on memory (e.g., emotionally intense events are stored more strongly).
- Emotional influence on decisions (e.g., risk tendency under fear vs confidence).

4. Multi-Agent Cognitive Controller

- Separate agents for memory, personality, emotion, and reasoning.
- Coordination loop:
input → perception → memory retrieval → emotional appraisal → personality filter → decision/response.
- Modular design so components can be evaluated or swapped independently.

5. Evaluation Sandbox

- A simple text-based interface to interact with the agent as it “lives” over multiple sessions.
- Logging of decisions, emotional states, and memory usage.
- Basic metrics/dashboard for:
 - consistency of personality
 - recall accuracy of past events
 - emotional appropriateness of responses

Existing Work (High-Level Overview)

Several lines of research are related to this project:

- **Memory-Augmented Language Models**
Recent works extend large language models with external memory to handle long-term context and continual learning. These systems can store and retrieve large histories of text, but they mostly focus on factual/contextual memory, not emotional or personality-linked memory.
- **Personality Modeling in AI Agents**
Some studies use LLMs to simulate personality traits or enforce a fixed persona. There are also early frameworks where personalities can evolve slightly over time in game-like simulations. However, personality is usually treated as a stylistic layer on top of conversation, not deeply integrated with memory and emotion.
- **Affective Computing and Emotional Agents**
Affective computing research demonstrates how to detect and model emotions and, more recently, how to use emotions to influence decision-making in agents. Yet, these emotional models are often isolated from rich long-term memory or detailed personality structures.
- **Multi-Agent and Cognitive Architectures**
Multi-agent frameworks and classic cognitive architectures (like “society of mind” style systems) show that decomposing cognition into specialized modules can be powerful. Some modern frameworks combine LLMs with multi-agent designs for social simulations or interactive environments, but they rarely bring together memory, emotion, and personality as core, persistent cognitive elements.

Gap (Research Opportunity)

From the existing literature, the following gaps emerge:

1. **Lack of Integration:**
Memory, personality, emotion, and decision-making are usually modeled separately. There is no widely adopted framework that combines all four into a single, coherent cognitive architecture.

2. **Limited Persistence & Evolution:**

Many systems simulate a persona or emotion only within a single session or short time window. Long-term continuity – where an agent accumulates experiences, updates its traits, and behaves consistently over weeks or months – is still underexplored.

3. **Emotion–Memory–Decision Link:**

Few systems explicitly link emotional state to what gets remembered, how it is recalled, and how it alters decisions. Human cognition is heavily influenced by affect; this connection is still weak in most AI agents.

4. **Evaluation of “Human-Like” Cognition:**

There is a lack of practical prototypes and evaluation frameworks that test an integrated cognitive system across memory recall, personality stability, emotional appropriateness, and decision behavior.

CognitiveOS aims to address these gaps by:

- proposing a modular architecture that links memory, personality, emotion, and reasoning;
- building a prototype that demonstrates persistent, evolving behavior; and
- designing small-scale experiments to evaluate human-likeness and stability.

Challenges (Technical & Research Risks)

1. **Scope Management**

- Human cognition is vast; the project must choose a realistic subset (text-based interactions, limited emotional categories, simple personality model) to complete within final-year timelines.

2. **Memory Design & Efficiency**

- Deciding what to store, how to summarize, and how to retrieve relevant information without the system becoming slow or noisy is non-trivial.

3. **Personality & Emotion Modeling**

- Selecting a suitable psychological model (e.g., Big Five for personality, simple valence–arousal for emotion) and mapping it meaningfully to agent behavior is a design challenge.

4. Evaluation Methodology

- Defining clear, measurable metrics for “personality consistency,” “emotional appropriateness,” and “human-like decision behavior” is difficult but necessary for a good research paper.

5. Data & Ethics

- If real personal data is used (e.g., conversations or reflections), privacy and consent must be carefully handled.
- The system should not claim to “be” a person; rather, it should be framed clearly as a simulation/model.

Conclusion

CognitiveOS represents an opportunity to explore an emerging frontier in artificial intelligence — the integration of long-term memory, personality traits, emotional processes, and decision-making within a unified multi-agent cognitive framework. While current research addresses these components individually, there is a clear gap in combining them into a cohesive system capable of persistent, human-like cognitive behavior.

This project aims to address that gap through a modular, interpretable architecture and a functional prototype that demonstrates how these cognitive elements can interact in meaningful ways. Beyond engineering value, the project also offers strong research potential: the architecture, experiments, and evaluation methods can contribute new insights to the fields of agentic AI, affective computing, and cognitive modeling.

With proper guidance and scope refinement, CognitiveOS has the potential to serve both as a solid final-year implementation project and as the foundation for a publishable research paper. It aligns closely with current AI trends while exploring an area that remains significantly underdeveloped, making it both academically relevant and practically impactful.

Referenced Articles and Links:

AI-powered personal knowledge / second brain:

- <https://irjaeh.com/index.php/journal/article/view/684>
- https://research-api.cbs.dk/ws/files/58846410/Liana_Razmerita_Personal_Knowledge_Management_open_archive.pdf
- <https://www.sciencedirect.com/science/article/pii/S0007681322000222>
- <https://journals.sagepub.com/doi/10.1177/20597991241287118>

Context-aware reminders / procrastination / nudges:

- <https://www.diva-portal.org/smash/get/diva2:831445/FULLTEXT01.pdf>
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC12022524/>
- <https://www.scitepress.org/Papers/2014/49607/49607.pdf>
- https://www.nber.org/system/files/working_papers/w20381/w20381.pdf

Second brain / tools (systems to cite):

- https://personal.asuswebstorage.com/feature_xbrain/?lang=en
- <https://get.mem.ai>

Smart personal assistants:

- <https://www.ijert.org/a-review-paper-on-smart-personal-assistant>
- <https://tijer.org/tijer/papers/TIJER2404248.pdf>
- <https://dl.acm.org/doi/abs/10.5555/1167533.1167535>